

Seven features, one decided stack.

Your feature map, checked against primary sources. **Three findings change the plan** — one of them unlocks the mechanic I previously said was impossible.

6 research tracks 26 claims verified 15 corrected 32 agents

Three findings that move the plan

1 · THE 40% POSTER SPLIT IS NATIVELY SUPPORTED. BETTER THAN BAGS.

I previously called this a showstopper — that no Solana primitive could route fees to a poster who never consented and has no wallet. **That was wrong.** Pump.fun ships an on-chain, permissionless **creator-fee-sharing** system: `create_fee_sharing_config` → `update_fee_shares_v2` → `distribute_creator_fees_v2`. It splits creator fees across **up to 10 arbitrary pubkeys** in basis points summing to exactly 10,000. No rule ties a shareholder to the creator identity — **any pubkey qualifies**.

Better still: `create_v2` takes `creator` as a plain Pubkey argument **independent of the signing user**. Insidor can pay for and sign a launch while assigning creatorship elsewhere. That single fact is the most valuable thing in the research.

2 · PHOTON AND AXIOM HAVE NO PUBLIC API. FEATURE 2 CANNOT START AS WRITTEN.

Not "hard to get" — **does not exist**. Axiom's own complete docs index (`llms.txt`, which enumerates every page they publish) contains zero developer, API, SDK or webhook entries. Photon's official GitBook has no developer content, and QuickNode's directory states flatly that Photon **"does not support API integrations or Telegram bots."** Both are Cloudflare-gated closed frontends.

There is no "integrate whichever is cheaper" decision to make, because there is nothing to price. The community "Axiom SDKs" require storing users' Axiom **passwords**, reading their email for login OTPs, and running headless Chrome past Cloudflare. Not shippable in a consumer product.

The replacement is strictly better on the axis you cared about: Photon charges your user 1% per buy *and* per sell and pays Insidor nothing. Jupiter lets you set your own integrator fee. You capture revenue from day one — which deletes the "build our own terminal later to take all fees" milestone.

3 · "SEARCH PF METADATA" IS NOT A THING YOU CAN CALL. YOU'RE BUILDING AN INDEXER.

The pump.fun SDK is **transaction-building only** — no search, no discovery, no metadata query. The endpoint that used to do ticker search, `frontend-api-v3.pump.fun/coins/search`, is **dead** as of this research. On-chain you get only a `uri` string; the image sits one HTTP hop deeper inside the IPFS JSON.

Match-by-image is a genuinely good idea — it fills the one hole a previous engineering review found that *nobody* had specced. But it is **not a 3-week feature**. It is a corpus you build and keep fresh.

Decided

LAYER	YOU CHOSE	VERDICT	SHIP
Coin creation	Pump.fun public SDK	Correct. <code>@pump-fun/pump-sdk</code> , MIT, official.	As planned
Fee routing	—	Natively supported. Up to 10 shareholders in bps.	With creation
Swap / trade	Photon or Axiom API	Impossible. No public API exists.	Jupiter instead
Charts	—	DexScreener embed — free, zero build.	Week 1
Trending coins	Axiom / Photon API	Same problem.	Jupiter / Birdeye
Wallet	Privy API	Correct. Pump.fun and Jupiter both run on it.	As planned
Coin search	PF SDK metadata search	No such API. Build an index.	Own indexer
Image match	PF SDK by image	Sound idea, wrong tool. Self-hosted embeddings.	Post-MVP
Comments	—	Supabase Realtime Broadcast + Postgres.	Weeks 1–3
Moderation	—	No vendor covers crypto harms. Build in-house.	With comments

Live feed

F1.1 Social layer integration

PRIORITY

Ingest → detect → cluster → score. The engine behind the whole feed.

SOURCE	X pay-per-use at \$0.005 per post read , hard ceiling 2M reads/month — a \$10,000/mo wall before Enterprise. New developers cannot buy Basic.
CONSTRAINT	That ceiling is ~46 posts/second. Thin for genuine viral detection — curate a watchlist of accounts rather than drinking the firehose.
RISK	X may terminate for any reason at sole discretion. Six InfoFi companies lost access in one day, 15 Jan 2026.

F1.3 Comments / share thoughts

PRIORITY

Users discuss a viral post inside Insidor.

BUILD	Supabase Realtime Broadcast (not <code>postgres_changes</code>) with Postgres as the store. Zero new vendors — you need Postgres anyway and Privy handles identity.
GOTCHA	Turn the Pro spend cap off on day one , or you're capped at 500 concurrent connections — precisely the viral spike the product exists to capture.
DON'T	Don't build your own websocket tier. At 50k DAU it saves under \$150/mo and costs two weeks plus permanent on-call.
COST	\$25/mo at 1k DAU → ~\$200/mo at 50k DAU.

THE MODERATION TRAP — THIS IS WHERE PRODUCTS IN THIS CATEGORY DIE

Every off-the-shelf moderation API — OpenAI, Perspective, Azure, Hive — classifies hate, violence and sexual content, and detects **exactly zero** of the harms that will actually appear: **contract-address spam, honeypot shilling, founder impersonation, coordinated pumps**. The crypto-native layer has to be built in-house.

Also: **Perspective API sunsets 31 Dec 2026** and closed quota increases in Feb 2026. Any plan naming it must be rewritten.

READ THIS BEFORE PUTTING A COMMENT BOX UNDER AN X POST

X's Display Requirements, verbatim: **"No other social or third party actions may be attached to a post. (e.g., subscribe, comment, like)."** Attaching an Insidor comment thread directly to a displayed X Post is a literal violation. Attach comments to the **narrative** — your own object — not to the rendered Post.

Thumbs up/down, name tags, revenue share for top-ranked users.

DESIGN AROUND THIS BEFORE YOU BUILD IT

Rev-share for commentary that promotes tokens Insidor earns trade fees on triggers **FTC Endorsement Guides** obligations (disclosure program, monitoring, penalties **\$51,744–\$53,088 per violation**) and **CFTC Rule 180.1** spot-manipulation exposure. It is also the exact shape X deleted the InfoFi category over.

The distinction that may save it: rewarding posting *on Insidor* is not rewarding posting *on X*. That distinction is ours, not X's, and it has never been tested. Keep it LATER — you were right to.

Live swap, chart, trade

Rewritten. Photon and Axiom are out; Jupiter is in and pays you.

F2 Jupiter Plugin + Ultra API

PRIORITY

3 WEEKS

Buy the coin without leaving the card — and take a cut.

```
// Week 1 — swap, one script tag, brings its own wallet layer
https://plugin.jup.ag/plugin-v1.js

window.Jupiter.init({
  displayMode: 'integrated', // embed directly in the feed
  enableWalletPassthrough: true, // reuse Privy — never connect twice
  formProps: { referralAccount, referralFee: 100 } // 100 bps to you
})
```

FEE	You set the integrator fee — max 255 bps . Jupiter takes ~20% of it on the aggregated <code>/order</code> path, 0% on the raw <code>/build</code> path.
EXECUTE	<code>api.jup.ag/ultra/v1/order</code> → <code>/execute</code> , with <code>referralAccount</code> + <code>referralFee</code> .
CHARTS	DexScreener iframe — <code>?embed=1&theme=dark&info=0&trades=0</code> . Free, zero build time.
TRENDING	<code>api.jup.ag/tokens/v2/toptrending/{5m 1h 6h 24h}</code> .
COST	Jupiter Developer \$25/mo (10 RPS). Free tier at 1 RPS genuinely works for a private beta.

CHECK THIS YOURSELF BEFORE LAUNCH

Jupiter's docs describe a **50 bps platform fee on tokens under 24 hours old** — which would be essentially every Insider trade. The live API returned **10 bps** on seconds-old tokens during verification. The docs and the live behaviour disagree. **Measure it on a real new-token quote before you price anything.**

DELETE A MILESTONE

"Build our own terminal later to take all the fees" was premised on renting a terminal that pays you nothing. Jupiter pays you from week one, so the milestone has no purpose. Spend that time on the indexer instead.

Creating coins

F3.1 Create — Pump.fun SDK

CORE

```
// @pump-fun/pump-sdk v1.36.0 (MIT)
// program 6EF8rrecthR5Dkzon8Nwu78hRvfCKubJ14M5uBEwF6P (mainnet + devnet)

PUMP_SDK.createV2Instruction({
  mint, name, symbol, uri,
  creator,          // ← ANY pubkey. independent of the signer.
  user,             // ← signs and pays
  mayhemMode: false, cashback: false
})

// limits: name ≤32, symbol ≤13, uri ≤200, creator ≠ Pubkey::default()
```

COST

\$0. Pump.fun charges nothing to create. Real cost is rent + tx fees (cents), plus your optional dev buy, plus 0.015 SOL from liquidity at graduation.

METADATA

`pump.fun/api/ipfs` was **alive and unauthenticated** at verification. Have a Pinata fallback — don't be single-homed on someone else's uploader.

```
// creator fee sharing — permissionless, on-chain
create_fee_sharing_config    // PDA seeds [b"sharing-config", mint]
update_fee_shares_v2        // set the split — SEE WARNING
transfer_creator_fees_to_pump_v2
distribute_creator_fees_v2   // permissionless, pro-rata, creates missing ATAs

// rules: max 10 shareholders · no duplicates · each > 0 · sum EXACTLY 10,000 bps
[ creator 5000 · poster 4000 · insidor 1000 ] // = 10,000 ✓
```

THE CONSTRAINT THAT SHAPES THE WHOLE FLOW

`update_fee_shares_v2` sets `admin_revoked = true` — **treat the split as set-once, at launch**. You cannot launch now and add the poster later when they show up.

The architecture that solves it: at launch time, deterministically derive a **Privy embedded wallet from the poster's social handle** and set *that* address as the shareholder. The wallet is addressable on-chain before the human has ever logged in, and

`distribute_creator_fees_v2` will happily pay a wallet nobody has claimed. The money sits there until they arrive.

Verify the exact revocation semantics against the live program before building on this — the fact-check flagged the consequence as possibly overstated.

TWO FOOTGUNS

1. Creating a `sharing_config` *permanently disables* `collect_creator_fee_v2` for that mint. If your distribution cranker breaks, there is no fallback collection path. Build and monitor it before you ship a single shared coin.

2. PDA seeds differ between programs: `[b"creator-vault", creator]` on Pump vs `[b"creator_vault", coin_creator]` on Pump AMM. **Hyphen versus underscore**. An easy and expensive bug.

F3.2 Find the existing coin — by ticker AND image

CORE

NOT 3 WEEKS

Fills the one gap a prior engineering review found nobody had specced — the buy path.

REALITY	No PF search API. <code>/coins/search</code> is dead. You are building an indexer.
CORPUS	Stream <code>create_v2</code> instructions for mint + name + symbol + uri. Hydrate images via Helius <code>getAssetBatch</code> — 1,000 mints/call, 10 credits/call . Use <code>content.files[].cdn_uri</code> ; never dereference IPFS yourself.
FILTER	Restrict to tokens with nonzero volume or bonding-curve position — 1–3M images, not 12M .
EMBED	Self-hosted SSOD on Modal L4. ~\$30–150 one-time for the backfill; ~\$1/mo ongoing for ~25k new tokens/day.
SEARCH	pgvector HNSW. Combine image similarity + fuzzy ticker + text embedding into one score.

CALIBRATE ON YOUR OWN DATA

Published near-duplicate thresholds come from **DISC21 — transformed Flickr photos**, not memes with text overlay and AI-generated logos. A tight crop of one face from a source image defeats global descriptors roughly half the time. **Hand-label a few hundred real post→token pairs and fit your own cutoff.** Don't ship a borrowed number.

Wallet, watchlist, search, trending

F4 Connect wallet — Privy

CORE

Stop researching this layer. **Pump.fun and Jupiter both run on Privy** — your competitors have load-tested the exact code paths you'll hit.

PACKAGE	<code>@privy-io/react-auth</code> . Solana is a first-class chain; full create / connect / sign / signAndSend / export surface.
CUSTODY	2-of-2 Shamir shares reconstructed only inside AWS Nitro Enclaves. SOC2 Type I+II, audited by Cure53 / Zelic / Doyensec. Defensible.
PRICE	0–499 MAU free · 500–2,499 \$299/mo · 2,500–9,999 \$499/mo . Includes 50k signatures and \$1M transaction volume per month.

THREE THINGS TO PRICE IN

The \$1M/month volume ceiling. 500 real traders averaging \$2,000/mo of volume hits it. Usage billing starts there — model it now, not when the invoice lands.

Privy's swap API pays you nothing on Solana (developer fees are cross-chain only) and gas sponsorship is mandatory for it. Route swaps through Jupiter, not Privy.

No card→memecoin. Funding is SOL or USDC, so every new user is onramp → SOL → swap: two flows, ~2.5% card fee plus swap. That's a real conversion cliff on the "buy the viral coin in one tap" promise.

F5 Watchlist — posts and coins

CORE

Two Postgres tables and a star button. The value is the **notification when a watched narrative finally mints** — that's the retention hook, not the list itself.

F6 Search — keyword and contract address

CORE

One query, both worlds: the post and the chart.

KEYWORD	Postgres full-text over your own narrative corpus + the token index you built for F3.2. Search is downstream of the indexer — it cannot ship before it.
BY CA	Resolve the mint through your index, then show the coin <i>and</i> the narrative it came from.

NARRATIVES

Your own scoring. Filter by age, views, velocity. This is the part nobody else has.

COINS

Not Axiom/Photon. **Jupiter** `toptrending` (free) or **Birdeye** (\$99–199/mo, adds websockets for live charts).

06 — SEQUENCE

What to build, in order

The three-week clock is real, but it applies to the trading terminal, not the indexer.

WEEK 0

Kill the assumptions

Three cheap checks before code. Quote a seconds-old token through Jupiter and **read the actual platform fee** (10 bps or 50 bps — docs and live behaviour disagree). Call `pump.fun/api/ipfs` and confirm it still accepts uploads. Read the live `update_fee_shares_v2` semantics on devnet to confirm exactly how one-shot the split is.

WEEK 1

Trading, in a day — then the corpus

Jupiter Plugin embedded in the feed with `enableWalletPassthrough` into Privy. DexScreener iframes for charts. Jupiter `toptrending` for coin data. **That is Feature 2 and half of Feature 7 done in week one.** Spend the rest of the week streaming `create_v2` instructions into your token index.

WEEK 2

Wallet, create, watchlist

Privy embedded wallets. Pump.fun `createV2Instruction` with the fee-sharing config wired to a poster-derived wallet. Watchlists for posts and coins.

WEEK 3

Comments, with moderation on day one

Supabase Realtime Broadcast, spend cap off. Ship the crypto-native filter **in the same release** — CA regex, impersonation checks, rate limits. Never after.

WEEKS 4–7

The indexer and image matching

Backfill and embed the filtered token corpus. Build the combined image + ticker + text scorer. **Hand-label your own evaluation set** and fit the threshold. Search (F6) ships on top of this.

LATER

Trenches, ranking, profiles, flex cards

As you had them. Take legal advice on rev-share *before* designing it, not before shipping it.

What it runs

LINE	BETA	1K DAU	50K DAU
X ingestion \$0.005/post read	~\$200	~\$1,600	up to \$10,000 (cap)
Jupiter Developer tier	\$0	\$25	\$25
Privy	\$0	\$299	\$499+
Supabase realtime + Postgres	\$25	\$25	~\$200
Birdeye optional, live charts	\$0	\$99	\$199
Helius indexing + RPC	\$0	~\$50	~\$250
Image embedding	\$30–150 once	~\$1	~\$1
Charts DexScreener	\$0	\$0	\$0
Monthly	~\$225	~\$2,100	\$1,200–11,000

X ingestion is the entire cost story. Everything else together is under \$1,200/mo even at 50k DAU. The 2M read/month ceiling is both your budget and your product limit — curate accounts, don't chase the firehose.

Verify these five

Adversarial fact-checking refuted 15 of 26 specific claims in this research — not the conclusions, but the exact endpoints, prices and method names. APIs in this space move weekly. Treat every code identifier here as needing confirmation on the day you use it.

- **The Jupiter new-token fee.** Docs say 50 bps under 24h; the live API returned 10 bps. Quote a real new token and measure. This decides your margin.
- **The fee-split revocation semantics.** How genuinely one-shot is `update_fee_shares_v2`? Test on devnet. It determines whether the poster wallet must be derived at launch.
- **Whether a shareholder wallet can be one nobody has claimed.** The whole 40% mechanic rests on paying an address before its human exists. Prove it on devnet with a fresh keypair.
- **X pay-per-use eligibility for a new developer account.** The 2M ceiling and the inability to buy Basic are the constraints that shape the product.
- **Whether Photon or Axiom have a private partner API.** Public: definitively no. Behind a BD email: unverified. One message answers it, and it would change Feature 2.

Three things to take advice on

- **Paying a poster who never consented.** Now technically easy, which makes the legal question the binding one. Right of publicity is state law with no federal statute, and consent is the primary defence. Ship the opt-out registry and the "not created or endorsed by" disclaimer *in the same release* as the split.
- **Rev-share for commentary.** FTC Endorsement Guides plus CFTC Rule 180.1 when you also earn fees on the thing being promoted. This is why it stays LATER.
- **Comments attached to X content.** The Display Requirements prohibition is explicit. Attach threads to your narrative object, never to the rendered Post.

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Supersedes sections 07 and 12 of the engineering architecture, which assumed Meteora/Bags and Jupiter-only.